

Nate Cirino

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Available May 2027

EDUCATION

Northeastern University

May 2027 | MGPA: 3.6/4.0

B.S. in Computer Science, Systems Concentration

Boston, MA

Coursework: Computer Systems, Digital Design & Computer Architecture, Computer Networks, Systems Security, Algorithms & Data Structures, Cybersecurity, Object-Oriented Design, Foundations of Software Engineering, Foundations of Data Science

EXPERIENCE

Wayfair

Jul 2026 – Present

Software Engineering Co-op

Boston, MA

- Own front-of-house device stack across all Wayfair physical retail brands (AllModern, Birch Lane, Joss & Main, Perigold): e-label hardware, mobile, and customer/associate-facing POS terminals
- Building a supplier-agnostic e-label API that abstracts 4 hardware suppliers behind one device interface, decoupling application software from per-vendor firmware and powering new outlet launches
- Shipped POS checkout flows integrating in-store scanner input: scan-to-basket customer switching and live associate-entry customer sync

Wolters Kluwer

Jul 2025 – Jun 2026

Software Engineer

Boston, MA

- Root-caused and overhauled Relay JMS broker serialization with json-io across 20+ services, eliminating timeouts at scale; contributed fixes upstream to the json-io open-source library
- Debugged and optimized Expert AI search and filtering across 20+ microservices, improving query relevance at scale
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing workflow, expanding GenAI clinical decision support to 250K+ clinicians
- Implemented OIDC/IdP authentication via Azure across 20 enterprise health systems, securing 10K+ clinicians

SculptAI

Jan 2025 – Aug 2026

Software Architect

New York, NY

- Maintained 99.99% uptime across 1-2M monthly model calls for 2K paying users via AWS WAF, Datadog, and K8s
- Led eng org restructure at \$1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows
- Architected Flask backend with RAG pipeline for personalized recommendations; continue as technical advisor

Chevron New Energies

May 2024 – Aug 2024

Software Engineer Intern

Houston, TX

- Implemented CI/CD workflows using Azure DevOps and K8s, enabling automated build, deployment, and rollback
- Built RESTful APIs and data pipelines using Azure Data Factory, improving product efficiency by 40%

PROJECTS

Kepler | Rust, C++, Linux, Raft, LSM-tree, MVCC, Tokio

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and LSM-tree engine, managing memory and on-disk layout manually with no garbage collector
- Engineered low-level storage internals on Linux: memtables, write-ahead log, leveled SSTables, and compaction with full SSTable rewrite every 4 flushes under inline, writer-blocking execution
- Implemented Raft election, log replication, and membership changes across concurrent async Tokio tasks, debugging race conditions and partition-tolerance failures in multi-node clusters
- Added MVCC transaction layer providing snapshot isolation and linearizable reads across the cluster

NOAA AUV Sonar Detection | Python, C, PyTorch, AWS Neuron SDK, SageMaker

April 2025 – July 2025

- Designed AWS Neuron SDK deployment path for onboard custom-silicon inference under hard real-time limits, targeting constrained compute on autonomous underwater vehicles
- Trained CNN models on raw thermal sensor spectrograms to detect novel aquatic signatures for NOAA AUVs
- Refined architecture and loss formulation to address class imbalance and hardware sensor noise, improving detection

Sentinel | FastAPI, Claude, pgvector, OpenTelemetry, Docker, Next.js

Apr. 2026 – Present

- Built an AI agent autonomously triaging production incidents via tool-use over Prometheus, Tempo, and Loki telemetry, cutting manual debugging time; running live in SculptAI production
- Built pgvector RAG over runbooks and postmortems with prompt caching on frozen schemas; 100% eval accuracy, with remediation gated behind human-in-the-loop approval

SKILLS & HONORS

Languages/Tools: C, C++, Python, Rust, Bash, Linux, Java, TypeScript, JavaScript, GDB, Docker, Kubernetes, Jenkins, Buildkite, Terraform, AWS, Azure, Groovy, PostgreSQL, GraphQL

AI-Enabled Engineer: Claude Code, Cursor, Devin, GitHub Copilot, CodeRabbit, Claude Cowork (agentic debugging, test automation, and code review)

Frameworks & Libraries: PyTorch, TensorFlow, Pandas, FastAPI, Flask, Node.js, Express, React, Next.js

Honors: Tech Lead at Northeastern Electronic Racing, json-io OSS Contributor, Dean's List (All Semesters), PawHacks